

Table of Contents

1. Executive Summary	1
The verdict in one paragraph	1
The five moves that matter most	1
2. Method and Evidence Base	2
3. Current-State Inventory	3
3.1 What is genuinely strong	4
3.2 What the platform is not	4
4. Gap Analysis: Eight Findings	4
F1. The assessment engine is skeletal (P0)	4
F2. The ISA spine stops before the opinion (P0)	5
F3. Arabic learners get videos, not a curriculum (P0)	5
F4. Progress is recorded, learning is not modeled (P1)	6
F5. Content depth is video-shaped, not text-shaped (P1)	6
F6. The Vercel database is a whiteboard (P1)	6
F7. Team features are ornamental for a workspace of one (P2)	6
F8. Discoverability is under-served by search (P2)	7
5. Improvement Roadmap	7
5.1 Tier P0 - the exam-readiness core	7
5.2 Tier P1 - retention and differentiation	8
5.3 Tier P2 - quality of life	8
6. Quick Wins	9
7. Success Metrics	9
8. Appendix: Survey Data	10

1. Executive Summary

AuditEdge Academy has grown into a genuinely unusual piece of software for an individual practitioner: a bilingual (English/Arabic, full RTL) learning workspace with 31 courses and 934 lessons, a 146-document standards library wired into a retrieval-augmented AI tutor, a 17-section risk-based audit program with PBC tracking, a findings register that evaluates misstatements against materiality, 20 sector risk profiles, and a voice layer that lets the tutor listen and speak in 23 voices including Egyptian Arabic. The engineering baseline is equally serious: a 912-check automated test battery, a self-provisioning deployment pipeline for Vercel, an offline-capable PWA shell, and a clean, verifiable release history across nineteen major versions.

This deep-dive was commissioned to answer one question: given where the platform stands today, which improvements would move exam readiness and audit craft the most? The analysis surveyed the live database, every user-facing view, all fourteen API route groups, the Prisma schema, the test suites and the deployment pipeline. It is grounded entirely in ground truth - every number quoted below was counted from the running system, not estimated from documentation.

The verdict in one paragraph

The platform's strength is concentrated in **reference and conversation**: standards content, sector intelligence, AI dialogue, and fieldwork tooling. Its weakness is concentrated in **assessment and retention**. Across 934 lessons there are exactly nine quizzes carrying roughly 48 multiple-choice questions, and 23 of the 31 courses - including every single Arabic-medium course - have no assessment at all. There is no question bank, no timed mock-exam mode, no spaced-repetition review, and no mastery model that could tell you which standard you are actually weak in. For a learner preparing for the SOXE / EEC exam or an ACCA-style audit paper, that is the single largest gap between what the platform offers and what exam preparation demands.



The five moves that matter most

The full roadmap in Chapter 5 contains nineteen initiatives across three priority tiers. Five of them dominate the value curve, and all five build on infrastructure that already exists in the codebase, so none requires a platform rewrite:

- **P0-1 Question Bank and Exam Simulation Center.** A seeded bank of 500-800 questions with standard tags, explanations and difficulty, plus a timed mock-exam mode that mirrors the SOXE / EEC

blueprint. This converts a content library into an exam weapon.

- **P0-2 Complete the ISA spine.** The reporting cluster (ISA 700/701/705/706), ISA 580, 550, 560, 600, ISQM 1 and the ethics code have no dedicated courses. These are precisely the standards Egyptian exam boards and ACCA-style papers over-sample.
- **P0-3 Spaced-repetition review engine.** A lightweight SM-2-style scheduler over lesson key points and missed quiz questions, surfaced as a daily review queue on the home screen.
- **P1-5 Case-based engagement simulation.** Turn the existing audit program into a guided, scored walkthrough from acceptance to the opinion - the single most differentiated feature the platform could add.
- **P1-7 Production data persistence.** On Vercel the database is rebuilt from a snapshot in a temporary directory on every deployment: progress, conversations and program data do not survive a redeploy. This is a data-loss bug wearing a deployment feature's clothes.

None of these is speculative. The course builder can already author the missing courses; the quiz player, attempt model and lesson key points already exist in the schema; the audit program already contains the engagement narrative a simulation would walk through. The roadmap chapters cite the exact models, routes and components each initiative touches, so effort estimates are grounded in the real codebase rather than generic guesses.

2. Method and Evidence Base

The analysis was performed against the running v19.2 workspace - the same build that ships to production - rather than against documentation, because documentation drifts and databases do not lie. Three evidence streams were combined. First, a structural survey of the codebase: all 37 React components under the audit feature folder, all 14 API route groups, the Prisma schema with its 13 models, the Zustand store and the i18n catalog were inventoried and cross-referenced. Second, a content census executed directly against the live SQLite database: course, module, lesson, quiz, question, material and usage counts, grouped by category, level, source platform and language. Third, the quality infrastructure was reviewed: the 912-check test battery, the browser smoke procedure, and the Vercel deployment pipeline including its self-provisioning snapshot mechanism.

Evidence stream	What was examined	Key outputs
Codebase survey	37 components, 14 API route groups, 13 Prisma models, store + i18n	Feature inventory (Ch. 3), roadmap touch-points (Ch. 5)
Content census	Live DB queries: courses, lessons, quizzes, questions, materials	Coverage stats (Ch. 4), appendix tables (Ch. 8)
Quality infrastructure	912-check suites, browser smoke, Vercel pipeline	Verification standard, deployment risk F7

Table 2-1. Evidence streams combined for this analysis.

One caveat deserves honesty: usage counters in the live database (lesson progress, quiz attempts, certificates) were reset to zero when the workspace database was rebuilt from the sanitized deployment snapshot during this session. No behavioral conclusions in this report rely on usage analytics; every finding is structural, drawn from what the platform contains and how it is built. Where a finding would benefit from usage data that does not yet exist - for example, which courses the learner actually opens - the report flags the missing metric as part of the finding itself.

Findings are graded by a simple rule: a gap only becomes a finding when it materially affects one of the platform's two jobs - preparing a senior associate for external-audit certification, and building the craft of executing engagements. Cosmetic observations were deliberately excluded, which is why this report contains eight findings rather than eighty.

3. Current-State Inventory

Before cataloguing gaps it is worth being precise about what already exists, because every roadmap initiative in Chapter 5 is deliberately designed to compound an existing strength rather than start from zero. The platform organizes into six capability areas, summarized in Table 3-1 and detailed in the subsections that follow.

Capability area	What exists today	Assessment
Curriculum	31 courses / 104 modules / 934 lessons; 8 in-house standards courses (ISA 240, 315, 330, 570, evidence, IFRS, Egyptian framework, analytics); 22 Arabic YouTube imports; 1 Coursera import; 879 lessons carry video	Strong breadth, uneven depth
Standards library	146 materials (41 ISA, 50 Egyptian standards, 55 IFRS) with extracted text content feeding the tutor's RAG index; source links to official texts	Differentiated asset
AI tutor	GLM 4.7-Flash / 4.6V vision / 4-Plus switcher; RAG + live web search; image attachments; 23-voice read-aloud with hands-free mic loop; searchable, exportable conversation history	Best-in-class for a personal tool
Engagement tooling	17-section risk-based audit program; PBC tracker with CSV export; findings/SAD register with ISA 450 materiality evaluation; sign-offs with tick-off export; analytical-procedure and ratio analyzers; KAM drafter; 20 sector risk profiles	Unique differentiator
Learning experience	Bilingual EN/AR with full RTL; dark mode; collapsible sidebars; PWA offline; XP, streaks, achievements, certificates	Polished shell
Engineering	Next.js 16 + Prisma; 912-check test battery; browser smoke procedure; self-provisioning Vercel snapshot deployment	Professional discipline

Table 3-1. Capability inventory of AuditEdge Academy v19.2.

3.1 What is genuinely strong

The engagement tooling deserves specific praise because it is the hardest thing in the product to copy. The audit program is not a static template: its findings register tracks every misstatement through open, passed and corrected states and evaluates the aggregate uncorrected amount against performance materiality and the clearly-trivial threshold exactly the way ISA 450 expects a senior to reason on a live file. The PBC tracker aggregates every document request across all 17 sections into one client-facing list with a CSV export, and the sign-off view enforces the ISA 230 discipline of preparer and reviewer dates. A junior associate who works this workflow honestly is rehearsing the real thing.

The AI tutor stack is the second standout. Retrieval over the standards library means answers cite the actual ISA and Egyptian standard texts rather than model memory; the vision model grades attached workpaper screenshots; and the voice layer - automatic answer reading, a hands-free conversation loop, Egyptian and Gulf Arabic neural voices - turns dead commute time into study time. The newly added follow-up chips (explain simpler, field example, quiz me, key points), clean regeneration, and Markdown export show a product being refined by someone who actually uses it daily.

3.2 What the platform is not

Honest framing also requires naming what the platform is not, because two common misconceptions waste improvement effort. First, it is not a learning-management system for a firm: it is a single-user workspace. The team view exists, but with one account it is a dashboard of one. Second, it is not a content-complete curriculum. The in-house courses cover the risk-assessment spine of the audit beautifully but stop there; the long tail of the catalog is imported video content with thin companion notes. Both facts sharpen the roadmap: invest in the single user's outcomes, and invest in the standards that are missing rather than adding more imported breadth.

4. Gap Analysis: Eight Findings

Each finding below states the evidence found in the system, why it matters for exam readiness or engagement craft, and the direction of the fix. Priority assignments (P0/P1/P2) carry into the roadmap in Chapter 5, where every finding reappears as one or more concrete initiatives with effort estimates.

F1. The assessment engine is skeletal (P0)

The entire platform carries nine quizzes containing about 48 questions, all multiple-choice with five to six items per quiz, attached to eight of the 31 courses. The 22 Arabic Academy courses and the Coursera import - 74 percent of the catalog - have no assessment at all, as Figure 4-1 shows. The Prisma schema stores questions as a JSON array on the Quiz row, which is serviceable for one-off course quizzes but cannot power a question bank: there is no question entity to tag with a standard, difficulty, or explanation, and no attempt record a review engine could learn from beyond the coarse QuizAttempt score.

Why it matters: exam boards do not test recognition, they test applied judgment under time pressure. Without a bank and a timed mode, the platform can teach but cannot rehearse. The fix - a Question model with standard tags, a seeded bank of 500-800 items, and an exam mode with section weighting - is specified as initiative P0-1 and is the highest-leverage work available.

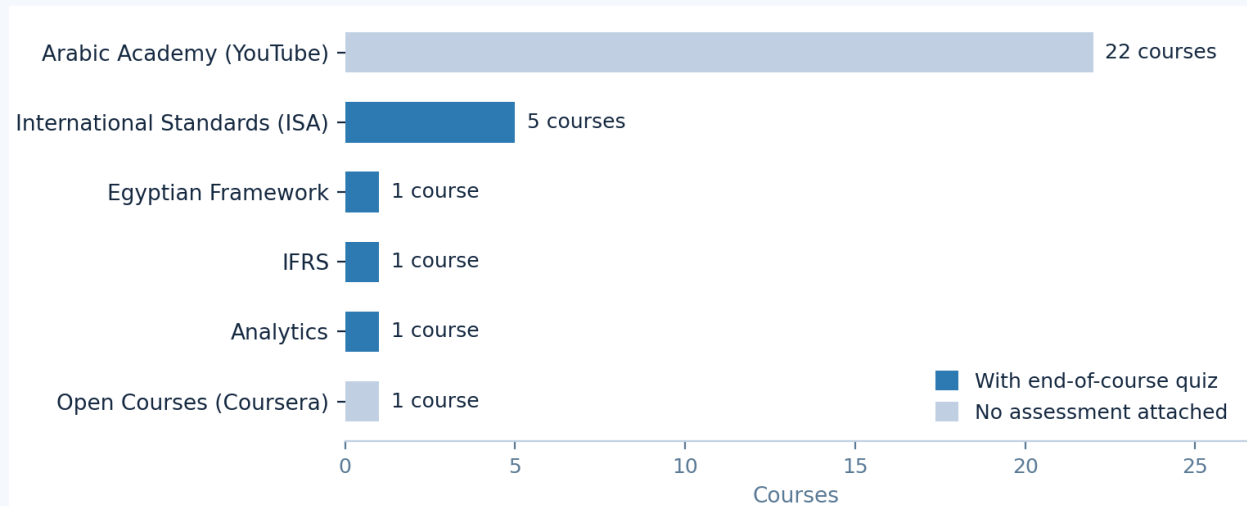


Figure 4-1. Assessment coverage by course category: only the eight in-house courses carry an end-of-course quiz.

F2. The ISA spine stops before the opinion (P0)

The in-house curriculum covers fraud responsibilities (ISA 240), risk assessment (315), responses (330), going concern (570), evidence and documentation (EVD-500), IFRS essentials, the Egyptian regulatory framework, and analytics. It then stops. Missing entirely: the reporting cluster (ISA 700, 701 KAM, 705 qualified opinions, 706 emphasis of matter), written representations (580), related parties (550), subsequent events (560), group audits (600), external confirmations (505), analytical procedures and sampling (520/530), quality management (ISQM 1), and the IESBA/Egyptian code of ethics. Some of these topics appear inside the audit program's methodology sections, but methodology checklists are not lessons - they cannot be studied, quizzed or tracked.

Why it matters: reporting is the most examined cluster in both the Egyptian SOXE/EEC syllabus and ACCA-style audit papers, and opinion modification scenarios are the classic exam differentiator. The course builder, lesson content schema and quiz tooling already exist, so this is a content-production task with light engineering - initiative P0-2.

F3. Arabic learners get videos, not a curriculum (P0)

All eight in-house courses - the only courses with structured lesson bodies, key points and quizzes - are written in English. The entire Arabic offering consists of imported YouTube playlists (plus one Coursera course), where the lesson 'content' is a short companion note averaging under a thousand characters around an embedded video. For a bilingual practitioner this is workable; for Arabic-medium exam preparation it

means the platform's deepest assets - the RAG library, the quizzes, the AI tutor's coaching prompts - are effectively English-only at the content layer. The UI itself is fully translated, which makes the asymmetry more visible, not less.

F4. Progress is recorded, learning is not modeled (P1)

The LessonProgress model records exactly one fact: that a lesson was completed at a timestamp. QuizAttempt records a percentage. Nothing in the system can answer the questions that actually drive improvement: which standards is the learner weak in, what is due for review today, and is exam readiness rising? There is no mastery model, no spaced-repetition scheduler, no review queue, and no weakness heatmap. The building blocks exist - every lesson already carries keyPoints suitable for flashcard fronts, and the tutor's system prompt already promises study planning - but nothing persists or schedules. Initiatives P0-3 and P1-4 close this.

F5. Content depth is video-shaped, not text-shaped (P1)

The mean lesson body is roughly 765 characters - about a third of a page - and eight lessons are effectively empty. That is a rational shape for companion notes to a video, and 94 percent of lessons do carry a video. But it caps the value of the two features that depend on rich text: the RAG index retrieves thin passages, and the read-aloud voice layer has little to read beyond what the tutor generates. Deepening the in-house courses' lesson bodies - worked examples, journal entries, mini-cases - would compound both the AI tutor and the future flashcard engine (initiative P1-6).

F6. The Vercel database is a whiteboard (P1)

The deployment pipeline is clever: a sanitized content snapshot is gunzipped into a temporary directory on the first request when the platform runs on Vercel's serverless filesystem. But that same mechanism means every redeploy wipes progress, conversations, quiz attempts and audit-program entries - anything the user created since the snapshot. For a personal learning tool whose whole promise is longitudinal (streaks, mastery, review queues), ephemeral storage is a structural contradiction. Initiative P1-7 specifies the migration to a managed Postgres (Neon or Supabase free tiers are sufficient for a single user) plus an export/import escape hatch.

F7. Team features are ornamental for a workspace of one (P2)

The team view, its pulse dashboard and the shared-progress model assume multiple users, but the platform is deliberately single-user with a self-healing workspace account. The features are not harmful, but they spend navigation space and maintenance effort on a fiction. The pragmatic options are either to embrace the single-user identity (rename the view to a personal analytics home and absorb the pulse widgets into it) or to defer any real multi-user work until a second real user exists. This report recommends the former and does not spend roadmap capacity on the latter.

F8. Discoverability is under-served by search (P2)

The catalog, library and tutor history each have their own local search, but there is no global search across courses, lessons, standards, sector profiles and conversations - the kind of command-palette (Ctrl+K) affordance that power users of a 934-lesson workspace reach for within a week. With the content census now exported, a lightweight client-side index over titles and key points is a one-day build (initiative P2-12).

5. Improvement Roadmap

The roadmap contains nineteen initiatives organized in three tiers. P0 items are the exam-readiness core and should be done first; P1 items compound retention and differentiation; P2 items are quality-of-life work that can interleave as energy allows. Effort is expressed in focused days for a single developer already familiar with the codebase, and every initiative names the models, routes and components it touches so the estimates can be checked against reality.

5.1 Tier P0 - the exam-readiness core

ID	Initiative	Builds on	Effort
P0-1	Question Bank + Exam Simulation Center: new Question and QuestionAttempt models (stem, options, answer, explanation, standard tag, difficulty, source); seed 500-800 items across ISA, Egyptian standards, IFRS and ethics; exam mode with 60/90-minute timer, section weighting per the SOXE / EEC blueprint, flag-and-review, and a results screen that feeds missed items into the review queue	Quiz player UI, QuizAttempt model, /api/quiz routes	6-8 days eng + seeding
P0-2	Complete the ISA spine: compact courses for ISA 700/701/705/706, ISA 580, 550, 560, 600, 505, 520/530, ISQM 1 and the ethics code - each with 3-5 lessons, worked examples and a 6-question quiz	Course builder (studio), lesson content JSON, quiz tooling	2-3 days eng + content
P0-3	Spaced-repetition review engine: SM-2-lite scheduler over lesson keyPoints and missed bank questions; ReviewItem model; daily review-queue card on Home with flashcard flip UI (EN/AR)	keyPoints on every lesson, home dashboard cards	3-5 days

Table 5-1. Tier P0 initiatives.

Sequencing within the tier matters. P0-1 should land first because its Question model is the dependency for both the review engine (P0-3 consumes missed questions) and the analytics layer (P1-4 aggregates attempt data by standard tag). P0-2 can proceed in parallel since it is mostly content production through the existing builder, and each new course quiz should be authored directly into the new bank so the bank and the curriculum grow together. If only one P0 item can be done this quarter, it should be P0-1: a platform with 800 exam questions and a timer changes what the product is.

5.2 Tier P1 - retention and differentiation

ID	Initiative	Builds on	Effort
P1-4	Mastery and coverage analytics: per-standard mastery score (quiz accuracy weighted by recency), weakness heatmap on the dashboard, readiness meter per exam section; AI study-plan generation persisted and tracked	QuestionAttempt + standard tags from P0-1	3-4 days
P1-5	Case-based engagement simulation: a guided walkthrough (client acceptance, risk assessment, response, completion, reporting) with branching decisions, seeded facts, and AI grading of each judgment; scores into achievements	17-section audit program, AI tutor, KAM drafter	5-7 days
P1-6	Arabic curriculum parity: Arabic lesson bodies and quizzes for the eight in-house courses (bilingual content field), Arabic question bank items; YouTube imports recast as supplementary	i18n layer, lesson content JSON, voice stack (ar-EG voices)	4-6 days + content
P1-7	Production data persistence: managed Postgres (Neon/Supabase) via DATABASE_URL for Vercel; snapshot mechanism kept as bootstrap fallback; user-data export/import endpoint as an escape hatch	Prisma datasource layer, /api/bootstrap	1-2 days
P1-8	Lesson depth pass: expand in-house lesson bodies to 1,500+ characters with worked examples and journal entries; fix the eight near-empty lessons	Lesson editor (studio)	ongoing, 30 min/lesson

Table 5-2. Tier P1 initiatives.

P1-5 deserves a note because it is the strategic bet. No commercial platform offers a scored, AI-graded engagement simulation grounded in a real 17-section program; the assets that make it possible here already exist. It is also the feature that would keep the platform valuable after the exams are passed, because it rehearses the craft of the job rather than the syllabus. P1-7 is the operational mirror of that bet: longitudinal features deserve a database that survives a Tuesday redeploy.

5.3 Tier P2 - quality of life

ID	Initiative	Effort
P2-9	Lesson notes and highlights with per-section ask-the-tutor buttons	3-4 days
P2-10	Workpaper template library: downloadable lead schedule, bank reconciliation, confirmations control sheet, going-concern memo (xlsx/docx)	2-3 days
P2-11	Podcast mode: batch-generate TTS audio per lesson for offline listening; download queue in the PWA	2-3 days
P2-12	Global command palette (Ctrl+K) across courses, lessons, standards, sectors and conversations	1-2 days
P2-13	CPE log: auto-accumulate cpeHours (already modeled) into an exportable certificate evidence log for SOXE renewals	1-2 days

ID	Initiative	Effort
P2-14	Team view rebrand as a personal analytics home; absorb pulse widgets	1 day

Table 5-3. Tier P2 initiatives.

6. Quick Wins

Independent of the tiers, six changes are each under a day of work and pay back immediately. They are listed in the order a busy week should tackle them, and none carries migration risk beyond a single additive Prisma field or pure front-end change.

- **Assessment badges on course cards.** Show a quiz-count chip on every course card in the catalog so the 23 assessment-free courses are visible at a glance rather than discovered after enrollment.
- **Continue-where-you-left-off card.** Persist the last-opened lesson and add a one-tap resume card to Home; the progress model already tracks completion, so this is a store field plus a card component.
- **One more quiz per in-house course.** Author a mid-course checkpoint quiz for each of the eight in-house courses through the existing studio tool - instantly doubles the assessment surface before the bank exists.
- **Fix the eight empty lessons.** The census identified eight lessons with near-empty bodies; a content pass on exactly those rows is a 30-minute task with the list in hand (see Appendix, Table 8-2).
- **Export the content census as a script.** Keep the survey script from this analysis in the repo under scripts/ and run it before each release to catch assessment-free courses and empty lessons automatically.
- **Deployment runbook note.** Add a two-line warning to the README deploy section: on Vercel, user data is ephemeral until P1-7 lands, so export before redeploying.

7. Success Metrics

A roadmap without measurement is a wish list. The platform's existing test culture makes verification cheap, so each tier gets a small set of metrics that can be computed from the database the initiatives themselves extend. Targets below are set for 90 days after the corresponding initiative ships.

Metric	Baseline (v19.2)	90-day target	Source
Courses with at least one assessment	8 of 31 (26%)	31 of 31 (100%)	Quiz table by course
Question bank size	48	500+	Question count
Mock exams attempted per month	0 (mode absent)	4+	ExamAttempt records
Median mock-exam score trend	n/a	rising quarter over quarter	ExamAttempt scores

Metric	Baseline (v19.2)	90-day target	Source
Review-queue items completed per active week	0 (engine absent)	25+	ReviewItem log
Per-standard mastery on priority standards	not measurable	70%+	mastery model (P1-4)
User-data durability on Vercel	wiped each deploy	survives deploys	P1-7 acceptance test

Table 7-1. Metrics and 90-day targets by initiative tier.

Two measurement rules keep these numbers honest. First, mastery and readiness metrics must be computed from attempts, never from lesson completion - the whole point of P1-4 is to stop conflating exposure with competence. Second, the content census script (quick win 5) becomes a release gate: any course shipping without an assessment, or any lesson shipping with an empty body, fails the pre-release checklist. What gets checked at release stops regressing.

8. Appendix: Survey Data

The tables below reproduce the raw census behind the analysis so future surveys can be compared apples-to-apples. All counts were taken from the live workspace database at v19.2; the survey script that generated them is kept in the repository under scripts/survey-analysis-v19.2.ts.

Dimension	Value
Courses	31 (8 in-house, 22 YouTube Arabic Academy, 1 Coursera)
Modules / lessons	104 / 934 (925 content lessons, 9 quiz lessons)
Lessons with video	879 of 934 (94%)
Mean lesson body length	about 765 characters; 8 lessons near-empty
Quizzes / questions	9 quizzes, about 48 MCQ items, 8 courses covered
Library materials	146 (41 ISA, 50 Egyptian standards, 55 IFRS)
Sector risk profiles	20
Audit program sections	17 (methodology, cycles, completion, reporting)
Course levels	6 Advanced, 17 Intermediate, 8 Foundation
In-house course language	English (8 of 8)
AI models	GLM-4.7-Flash (default), GLM-4.6V-Flash (vision), GLM-4-Plus
Voices	23 (16 Edge neural incl. ar-EG / Gulf Arabic, 7 Z.ai)
Test battery	912 checks (sectors 515+351, engagement 28, models 14, edge-tts 4)

Table 8-1. Content and platform census, v19.2.

Category	Courses	With quiz	Notes
International Standards (ISA)	5	5	ISA 240, 315, 330, 570, EVD-500
Arabic Academy (YouTube)	22	0	Arabic-medium video imports
IFRS	1	1	IFRS Essentials for External Auditors
Egyptian Framework	1	1	Regulatory and professional framework
Analytics	1	1	Analytics-Driven Auditing
Open Courses (Coursera)	1	0	Excel financial modeling import

Table 8-2. Course categories and assessment coverage.

A closing observation on method: this census took under an hour to produce because the platform's own tooling - Prisma, the seed scripts and the test battery - made the data legible. That is the quiet advantage of the engineering discipline this platform already practices, and it is why every roadmap initiative here can be verified the day it lands.